

Advanced EDA and Feature Engineering

Data Cleaning, Outlier Treatment and Predictive Feature Design

PROGRAM	Batch 2026
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1 OBJECTIVE

This project performs an end-to-end exploratory data analysis (EDA) and feature engineering pass on the raw e-commerce order dataset (Dataset_for_Data_Analytics.xlsx), preparing a clean, model-ready dataset for downstream projects (fraud detection and customer segmentation).

Requirements Addressed

- Handle missing data via statistical imputation (mean, median, or KNN).
- Identify and neutralize outliers using IQR and Z-score methods.
- Engineer at least three new predictive features.
- Apply Pandas, NumPy, statistical analysis, data cleaning, and feature extraction techniques.

AT A GLANCE

- Two imputation strategies applied: median for standard fills, KNN (k = 5) demonstrated for similarity-based imputation.
- Four numeric columns screened for outliers; all extreme values capped via IQR-based Winsorization.
- Eight new predictive features engineered, feeding directly into Projects 2 and 3.

2 METHODOLOGY

2.1 Missing Data Handling

Categorical missing values in CouponCode were filled with the label 'None'. Numerical columns with missing values were imputed using the column median, which is robust to skew and outliers. A KNN imputation demonstration (k = 5, on standardized Quantity, UnitPrice, ItemsInCart, and TotalPrice) was also run to illustrate a more advanced, similarity-based imputation approach.

2.2 Outlier Detection and Neutralization

Outliers in Quantity, UnitPrice, TotalPrice, and ItemsInCart were identified using the IQR method (1.5× IQR fences) and neutralized via Winsorization, where values were capped at the lower and upper bounds rather than removed. This preserves sample size while limiting the influence of extreme values.

2.3 Feature Engineering

Eight new features were engineered from the base order data:

- **DayOfWeek** — calendar day name derived from the order date.
- **IsWeekend** — binary flag for Saturday or Sunday orders.
- **OrderMonth & OrderYear** — seasonality-related date parts.
- **IsDelivered** — binary flag for successfully delivered orders.
- **HasDiscount** — binary flag for coupon usage.
- **AvgItemValue** — TotalPrice divided by ItemsInCart.
- **DiscountCode** — normalized coupon category (SAVE10, WINTER15, FREESHIP, Other, or None).

2.4 Encoding, Multicollinearity, and Scaling

Categorical columns (Product, PaymentMethod, OrderStatus, ReferralSource, DayOfWeek, DiscountCode) were one-hot encoded with drop_first set to true, avoiding the dummy variable trap. A correlation scan (threshold of absolute r greater than 0.8) checked for multicollinearity among the resulting numeric columns, dropping the lower-variance member of any highly correlated pair. Continuous numeric columns were then standardized (z-score) via StandardScaler, added as new scaled columns alongside the originals.

3 TOOLS AND TECHNIQUES

Category	Techniques Used
Imputation	Median imputation, KNN Imputer (k = 5)
Outlier handling	IQR fences, Winsorization (capping)
Feature engineering	Date decomposition, ratio features, categorical flags
Encoding	One-hot encoding (drop_first)
Diagnostics	Pearson correlation matrix, skewness
Scaling	StandardScaler (z-score)
Core libraries	pandas, NumPy, SciPy, scikit-learn

4 OUTPUT

The cleaned, encoded, and scaled dataset was exported as a timestamped CSV file (output/processed_data_[timestamp].csv) and used as the input for Project 2 (fraud detection) and Project 3 (customer segmentation), provided in this deliverable as Project1-Output.csv.

5 KEY TAKEAWAYS

- Missing values were fully resolved, with no nulls remaining in the final dataset.
- All four numeric columns checked for outliers were capped rather than dropped, preserving the full order history.
- Eight engineered features add time-based, behavioral, and value-based signal for the downstream classification and clustering projects.
- Multicollinearity screening kept the encoded feature set lean before scaling.